

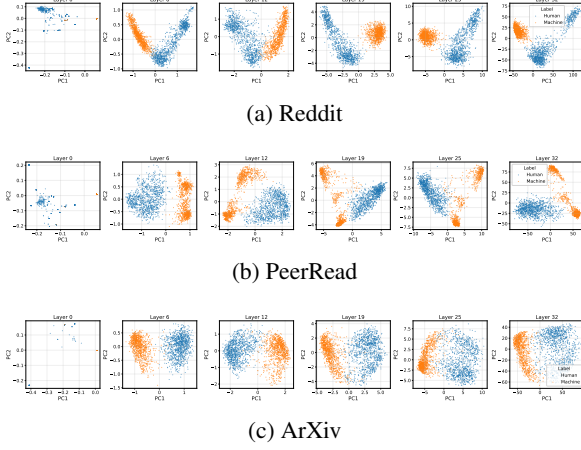


Figure 7: Projections of human- and machine-generated text hidden states onto the first two principal components across layers. This figure extends Figure 1 to the Reddit, PeerRead, and ArXiv subsets of M4GT (Wang et al., 2024).

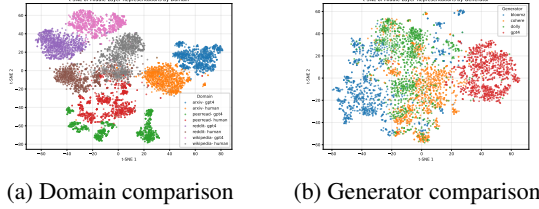


Figure 8: Low-dimensional t-SNE projections of human- and machine-written text from different domains (a) and generators (b) in the M4GT dataset (Wang et al., 2024).

hence does not preserve linear structure, the observed clustering is consistent with our earlier finding that human and machine representations are linearly separable within individual domains and generators.

In Figure 3, we analyzed the four representation quality metrics on the Wikipedia subset. Figure 9 extends this analysis to the ArXiv, Reddit, and PeerRead subsets of M4GT (Wang et al., 2024), as well as selected subsets from DetectRL (Wu et al., 2024), RAID (Dugan et al., 2024), MultiSocial (Macko et al., 2025), and TSM (Quaremba et al., 2026). Across these diverse settings, we observe the same qualitative pattern as for Wikipedia: MGT representations exhibit lower entropy, effective rank, and intrinsic dimensionality, while showing higher anisotropy than HWT representations. Although the differences are sometimes less pronounced, particularly in datasets that include adversarial attacks or multiple generators, the overall trend remains consistent.

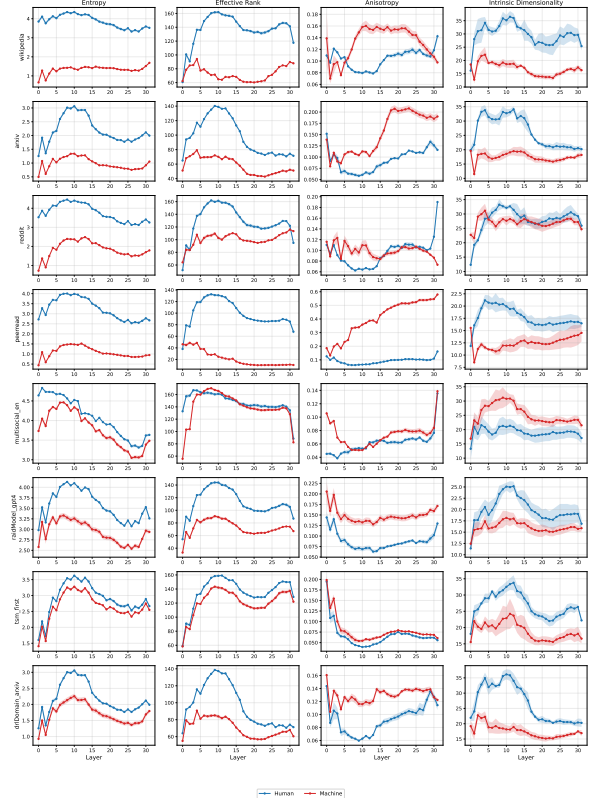


Figure 9: Representation-quality metrics for human and machine latent representations from the perspectives of information content (Entropy, Effective Rank) and geometric structure (Anisotropy, Intrinsic Dimensionality). Extended results to other domains than Wikipedia from M4GT (Wang et al., 2024) and selected subsets from DetectRL (Wu et al., 2024), RAID (Dugan et al., 2024), MultiSocial (Macko et al., 2025), and TSM (Quaremba et al., 2026).

B Experimental Setup

B.1 Benchmarks

We briefly introduce each benchmark and describe how the data were sampled and split. While we use M4GT (Wang et al., 2024) for the representation analysis in Section 3, initial experiments on its domain subsets revealed saturated results, with several baselines achieving near-perfect performance. We therefore include more challenging benchmarks for the detection experiments.

- **DetectRL** (Wu et al., 2024): improves upon prior benchmarks by constructing data that more closely reflects real-world detection scenarios, including a range of adversarial attacks such as prompt-based and paraphrase attacks.⁹ We use the Task 2 domain dataset and randomly sample training and test sets. Conse-

⁹<https://github.com/NLP2CT/DetectRL>

quently, each domain split contains examples generated by different LLMs and subjected to diverse attack strategies.

- **MultiSocial** (Macko et al., 2025): focuses on social media text, which is typically shorter, noisier, and more linguistically diverse than text from traditional domains. The benchmark covers 22 languages across 5 social media platforms.¹⁰ We randomly sample training and test sets from the full dataset, resulting in a diverse mixture of languages, text lengths, writing styles, and platforms.
- **RAID** (Dugan et al., 2024): is one of the largest MGT detection benchmarks, containing over 10 million text samples generated across diverse models, domains, decoding strategies, and adversarial attacks.¹¹ We construct generator-specific subsets by randomly sampling from the full dataset. As a result, each subset contains a diverse mixture of domains, decoding strategies, and attack types.
- **TSM** (Quaremba et al., 2026): focuses on the Wikipedia domain and text-generation tasks that closely resemble real-world editing behavior, including first-paragraph generation, paragraph extension, summarization, and text style transfer.¹² We use the English split and randomly sample training and test instances within each task across six generators.

For the experiments on detecting varying degrees of AI-edited texts in Section 5.6, we use the following benchmarks:

- **APT-Eval** (Saha and Feizi, 2025): APT-Eval contains 15k AI-polished texts from diverse domains, covering varying degrees of AI involvement in human-written text.¹³ The benchmark evaluates five detectors and quantifies edit strength using semantic cosine similarity, Jaccard distance, and Levenshtein distance. We use these metrics to correlate MGT probe projection scores with the degree of AI editing. To obtain a balanced sample, we evenly draw 200 instances from each polishing category: (1) extremely minor, (2) minor, (3) slightly major, and (4) major.

¹⁰<https://github.com/kinit-sk/multisocial>

¹¹<https://github.com/liamdugan/raid>

¹²https://github.com/gerritq/tsm_bench

¹³<https://github.com/ShoumikSaha/ai-polished-text>

- **EditLens** (Thai et al., 2026): EditLens contains AI-edited human-written text from four domains: reviews, creative writing, general educational web articles, and news articles.¹⁴ The benchmark uses three LLMs to edit human-written text with varying levels of intervention. We randomly sample from the full dataset using the same procedure as for APT-Eval.

B.2 Baseline Detectors

We test 16 MGT detectors covering zero-shot and supervised methods. Below we briefly introduce each detector and implementation details. If not otherwise stated, we implement each detector with Llama-3-8B (Grattafiori et al., 2024) for fair comparison.

Zero-shot

- **Likelihood** (Solaiman et al., 2019): A simple baseline that computes the average token log-likelihood of a text. Higher log-likelihood indicates that the text is more likely to have been generated by an LLM.
- **LLR** (Su et al., 2023): The Log-Likelihood Log-Rank Ratio (LLR) combines token log-likelihood and token rank. Log-likelihood captures a model’s absolute confidence in the observed token, while rank captures its relative confidence. Higher LLR scores indicate that a text is more likely to be machine-generated.
- **Rank** (Gehrmann et al., 2019): Assigns each token its rank under the language model’s predicted token distribution. A lower average rank indicates that the observed tokens are more predictable and therefore more likely to have been generated by an LLM.
- **Revise** (Zhu et al., 2023b): A rewrite-based detector motivated by the observation that LLM-generated text changes less under rewriting than human-written text, as it already conforms closely to the statistical patterns learned by the model. Higher similarity between the original and rewritten text therefore indicates a higher likelihood of machine generation. Instead of ChatGPT, we use LLaMA-8B-Instruct¹⁵ for rewriting.

¹⁴<https://github.com/pangramlabs/EditLens>

¹⁵<https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>

- **Binoculars** (Hans et al., 2024): Uses two LLMs to compute the ratio of perplexity to cross-perplexity. Cross-perplexity measures how surprising the token predictions of one model ("observer") are to another model ("performer"). Lower scores indicate that a text is more likely to be machine-generated.
- **FastDetectGPT** (Bao et al., 2023): An efficient variant of DetectGPT (Mitchell et al., 2023). DetectGPT exploits the observation that LLM-generated text tends to lie in regions of negative curvature of the model’s log-probability surface. FastDetectGPT approximates this signal more efficiently through sampling-based estimation.
- **GECScore** (Wu et al., 2025b): Computes the Grammar Error Correction Score (GEC-Score), motivated by the observation that human-written text typically contains more grammatical errors than LLM-generated text. Following the original method, the score is derived from the similarity between the original text and its grammar-corrected version. Due to computational constraints, we use the 8B model variant instead of LLaMA-3-70B.
- **RAIDAR** (Mao et al., 2024): Another rewrite-based detector motivated by the observation that LLMs tend to modify human-written text more substantially than machine-generated text. The method derives an MGT signal by measuring the edit distance between the original text and its rewritten version.
- **RADAR** (Hu et al., 2023): Trains a detector through adversarial learning between a paraphraser and a discriminator, using Vicuna-7B (Zheng et al., 2023) as the base model.
- **BiScope** (Guo et al., 2024): Motivated by the observation that LLMs tend to memorize local context differently for human- and machine-generated text. The method computes cross-entropy losses between output logits and both the ground-truth token and the immediately preceding token. A classifier is then trained on statistics derived from these losses.
- **TextFluoroscopy** (Yu et al., 2024): Argues that the strongest MGT signals emerge in middle layers. The method identifies the middle-layer representation whose vocabulary-space distribution differs most from the first and last layers and trains a nonlinear MLP classifier on that representation. We replace the originally used GTE-Qwen1.5-7B-Instruct model¹⁷ with LLaMA-3-8B.
- **RepreGuard** (Chen et al., 2025): Identifies a probing direction by computing the first principal component of the activation differences between human- and machine-generated text. Although RepreGuard is training-free, it still requires labeled examples to estimate the probing direction through PCA.
- **EditLens** (Thai et al., 2026): Trains a regression model to estimate the degree of AI editing in a text. We use the released RoBERTa-Large model,¹⁸ which was fine-tuned on data spanning different levels of AI-edit strength.
- **RoBERTa** (Liu et al., 2019): We fully fine-tune RoBERTa-Base¹⁹ on DetectRL, TSM, and RAID. For MultiSocial, we instead fine-tune XLM-RoBERTa-Base²⁰ to support multilingual inputs. We use a batch size of 32, a learning rate of 2×10^{-5} , weight decay of 0.01, and train for two epochs.

Supervised detector

- **OpenAI-RoBERTa** (Solaiman et al., 2019): A RoBERTa (Liu et al., 2019) classifier fine-tuned to distinguish GPT-2-generated text¹⁶ from human-written WebText data.
- **ID** (Hu et al., 2023): Uses the Persistent Homology Dimension (PHD) estimator (Schweinhart, 2021) to measure the intrinsic dimensionality of text representations. The resulting scalar feature is fed into a logistic regression classifier. We use XLM-RoBERTa (Conneau et al., 2020) as the backbone model.

¹⁶<https://huggingface.co/openai-community/gpt2-xl>

¹⁷<https://huggingface.co/Alibaba-NLP/gte-Qwen1.5-7B-instruct>

¹⁸https://huggingface.co/pangram/editlens_roberta-large

¹⁹<https://huggingface.co/FacebookAI/roberta-base>

²⁰<https://huggingface.co/FacebookAI/xlm-roberta-base>